

SHAHAD ALSANEEA

Software Engineer | Full-Stack & System Foundations

Riyadh, Saudi Arabia | +966506421412 | alsanee.shahad@gmail.com

PROFESSIONAL SUMMARY

A highly analytical and solution-driven Software Engineer with a solid foundation in computer science core principles, low-level system concepts, and end-to-end full-stack architectures. Graduate of the intensive, project-driven Holberton School software engineering program. Proficient in multiple programming paradigms, including lower-level memory management (C) and high-level web environments (Python, React, Django). Adept at applying strict mathematical analysis (Big O) to refine sorting mechanisms and complex data structures. Demonstrated ability to resolve code merging conflicts under scale, implement microservices pipelines, and bridge software architecture with data intelligence assets to optimize product performance.

TECHNICAL SKILLS

- **Core Programming Languages:** Python, C, JavaScript (ES6+), TypeScript, SQL, Bash Scripting.
- **Software Architecture & OOP:** Object-Oriented Programming (OOP), MVC Pattern, Test-Driven Development (TDD), System Design Patterns, UML Modeling.
- **Full-Stack Web Engineering:** Django REST Framework, React.js, Flask, Node.js, WebSockets, HTML5, CSS3, RESTful APIs, Asynchronous Programming.
- **Data Structures & Algorithms:** Binary Trees, Hash Tables, Linked Lists, Advanced Sorting Algorithms, Complexity Analysis (Big O Notation).
- **DevOps & Infrastructure:** Docker, Linux/Unix Command Line, Nginx, Shell Input/Output Redirection, Git & Version Control (Complex Branching/Merging).
- **Databases & Storage Engines:** PostgreSQL, MySQL, SQLAlchemy ORM, Database Schema Optimization, NoSQL Foundations.
- **Data Intelligence Integration:** ETL Processing Rules, Custom Analytics Dashboards, Power BI Dashboard Architecture, Technical Metrics Extraction.

KEY TECHNICAL PROJECTS

Rauf – Full-Stack Tele-Veterinary Platform | Software Engineer

- Coordinated end-to-end lifecycle development of a multi-tier clinical platform, mapping complex healthcare domain logic into robust architectural components.
- Designed and implemented secure JSON API layers with Django REST Framework, incorporating rigorous payload validation, entity pagination, and route access control.
- Deployed asynchronous real-time infrastructure utilizing WebSockets and Redis channel layers to synchronize media session state variables smoothly.
- Engineered intuitive frontend modules in React, managing centralized state synchronization, view token routing, and secure user workflow paths.

HBnB (AirBnB Clone) – Modular Full-Stack Framework / *Software Engineer*

- Designed and documented an extensible OOP application model blueprint using detailed UML object, class, and interaction diagrams.
- Constructed a pluggable backend storage abstraction paradigm with support for both flat filesystem serialization and dynamic MySQL relational engines utilizing SQLAlchemy ORM.
- Built out and tested complete RESTful APIs with Flask under strict Test-Driven Development (TDD) pipelines, keeping compliance to zero regression tolerances.

Custom UNIX Command Interpreter (Simple Shell) / *Systems Programmer (C)*

- Built a POSIX-compliant execution environment from scratch in C, strictly micro-managing memory states (``malloc`/`free``) to isolate and mitigate execution runtime leaks.
- Implemented standard operational process isolation using Unix kernel calls (``fork``, ``execve``), multi-tier string parsers, and custom-mapped data storage indexes (``Hash Tables``, ``Linked Lists``).

Dynamic Log Tracking Site & Analytics Dashboard / *Full-Stack Creator*

- Programmed a custom event logging portal that processes platform metric payloads to monitor visitor telemetry in real time.
- Created an integrated dashboard framework to map relational data directly onto key architectural health and behavioral KPIs.

EDUCATION & TRAINING

Intensive Software Engineering Program

Holberton School at Tuwaiq Academy | Riyadh, Saudi Arabia | Graduate

- Immersed in a project-driven curriculum mimicking strict corporate engineering cycles, focusing heavily on baseline logic, lower-level compilation, and distributed computing.
- Core Areas: Advanced Computer Science algorithms, object-oriented concepts, system scripting, server environment routing, and automated cluster deployment using Docker.

Bachelor of Dental Surgery (BDS)

Riyadh Elm University | Riyadh, Saudi Arabia

PROFESSIONAL & CORPORATE EXPERIENCE

Power BI Developer – Corporate Simulation

PwC Switzerland (via Forage) | 2024

- Engineered semantic visual layouts to represent complex business datasets, translating foundational multi-tiered enterprise metrics into high-level solutions.

General Dentist

Almeswak Clinic | Riyadh, Saudi Arabia | 2020 – 2023

- Algorithmic Diagnostics: Evaluated clinical edge cases and structured step-by-step physical diagnosis pathways, matching the logical procedures applied during rigorous server error-handling routing.
- Systematic Data Quality: Managed dynamic and compliant electronic patient asset databases, preserving strict data schema alignment and zero-tolerance error tracking rules.

PROFESSIONAL CERTIFICATIONS

- Microsoft Certified: Power BI Data Analyst Associate (PL-300 - Enterprise Data Modeling)
- Google Advanced Data Analytics Professional Certificate (Statistical Modeling & Python Optimization)
- Google Business Intelligence Professional Certificate (ETL Processes & Asset Pipeline Design)
- Google Data Analytics Professional Certificate (SQL Queries & Relational Optimization)
- Business Intelligence & Data Analyst (BIDA®) | Corporate Finance Institute (CFI)